

Rafi Ibn Sultan

Multimodal AI Researcher | Vision-Language Models | Grounded Reasoning
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Summary

PhD candidate in Computer Science (expected August 2027) developing multimodal AI systems whose reasoning and predictions remain grounded in visual evidence. Research spans large vision-language models, spatial reasoning, visual grounding, segmentation, and medical image analysis, with first-author work at CVPR, BMVC, ECAI, and Medical Physics. Lead projects from problem formulation and model development through post-training, evaluation, publication, and release of research artifacts.

Technical Skills

Programming and ML: Python, PyTorch, CUDA, Linux, Git, Hugging Face, MONAI, OpenCV, scikit-learn, Weights & Biases

Models and Training: Large vision-language models, vision transformers, SAM, supervised fine-tuning, GRPO, LoRA/PEFT, contrastive learning, distributed multi-GPU training

Research Areas: Multimodal reasoning, visual grounding, spatial reasoning, semantic segmentation, depth reasoning, representation learning, 3D medical imaging

Research Experience

Graduate Research Assistant | Wayne State University
Detroit, MI

Sep. 2023–Present

- Lead multimodal AI projects from problem formulation through model implementation, distributed training, ablation design, evaluation, publication, and release of research artifacts.
- Develop and post-train large vision-language models using supervised fine-tuning and reinforcement learning, with emphasis on grounded spatial reasoning and interpretable intermediate predictions.
- Lead a clinical collaboration with Henry Ford Health on coronary artery segmentation from non-contrast CT, translating clinical requirements into model design and clinically meaningful evaluation.
- Mentor junior PhD researchers on implementation, experiment design, analysis, and technical writing while coordinating work across computer vision, machine learning, and domain collaborators.

Selected Research Projects

WalkGPT: Grounded Vision-Language Navigation

CVPR 2026

- Built a grounded conversational vision-language system that combines language generation, semantic segmentation, and depth reasoning to provide visually anchored pedestrian-navigation guidance.
- Developed the PAVE benchmark with approximately 41K question-answer pairs and evaluated real-world transfer on 701 SANPO-Real sessions; prepared the dataset, model weights, project site, and demonstration materials for public release.

SpatialCORE: Confidence-Guided LVLM Post-Training

Under review, 2026

- Post-trained Qwen3-VL-8B-Thinking with GRPO using confidence-weighted grounding rewards, Hungarian matching, and uncertainty-aware supervision derived from automatically generated spatial evidence.
- Designed an end-to-end pseudo-grounding pipeline using GPT-4o phrase extraction and Grounding DINO, with confidence calibration and manual quality auditing to support scalable spatial supervision.

MedPlex: Vision-Language Medical Segmentation

BMVC 2026

- Designed bidirectional multi-stage fusion and concept-grounded contrastive alignment to continuously co-adapt visual and textual encoders for 3D CT and MR segmentation.
- Evaluated across diverse multi-organ, brain, cardiac, and free-text abnormality benchmarks, including transfer and textual-consistency studies.

GeoSAM: Multimodal Prompting for Geospatial Segmentation

ECAI 2025

- Adapted Segment Anything with multimodal prompts for mobility-infrastructure segmentation and cross-city generalization.
- Developed and evaluated a multimodal prompting strategy in Washington, D.C., then demonstrated transfer to an unseen city and urban environment in Cambridge, MA.

Selected Research Projects (continued)

3D Coronary Artery Segmentation

Medical Physics, 2026

- Developed a neighborhood-attention transformer for left anterior descending artery segmentation from non-contrast planning CT, using large-scale CCTA pretraining followed by clinical evaluation.
- Built a two-stage research pipeline using 1,000 public CCTA volumes for pretraining and a private clinical cohort for five-fold evaluation in collaboration with medical physicists and clinicians.

Teaching and Mentoring Experience

Graduate Teaching Assistant | Wayne State University
Detroit, MI

Aug. 2022–Aug. 2023

- Led Software Engineering labs and supervised 5–6 teams of 4–5 students per term through requirements, implementation, testing, and final delivery.

Lecturer | Varendra University
Rajshahi, Bangladesh

Oct. 2019–Aug. 2022

- Served as instructor of record for Microprocessor and Assembly Language, Introduction to Object-Oriented Programming, and Computer Fundamentals, teaching 30–40 students per course with responsibility for lectures, assessments, and grading.
- Supervised approximately eight undergraduate thesis and capstone projects from proposal development through implementation, evaluation, and defense.

Education

PhD, Computer Science, Wayne State University, Detroit, MI
Advisor: Prof. Dongxiao Zhu

Expected Aug. 2027

MSc, Computer Science, Wayne State University, Detroit, MI

Dec. 2025

BSc, Computer Science and Engineering, RUET, Bangladesh

Nov. 2018

Publications

First-Author Publications

1. **RI Sultan**, H Zhu, X Zhou, C Li, P Khanduri, M Brocanelli, D Zhu. “WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation.” *CVPR*, 2026.
2. **RI Sultan**, C Li, Y Demetriou, AI Ghanem, JP Kim, J Cunningham, H Bagher-Ebadian, D Zhu, KS Thind. “A Neighborhood Attention Transformer Network for Enhanced 3D Segmentation of the Left Anterior Descending Artery.” *Medical Physics*, 2026.
3. **RI Sultan**, H Zhu, C Li, D Zhu. “MedPlex: Deep Vision-Language Co-Adaptation for Clinically Grounded Medical Segmentation.” *BMVC*, 2026.
4. **RI Sultan**, C Li, H Zhu, P Khanduri, M Brocanelli, D Zhu. “GeoSAM: Fine-Tuning SAM with Multi-Modal Prompts for Mobility Infrastructure Segmentation.” *ECAI*, 2025.
5. **RI Sultan**, X Zhou, MSA Chowdhury, C Li, P Khanduri, M Brocanelli, D Zhu. “SpatialCORE: Spatially Confident Reasoning for Large Vision-Language Models.” Under review, 2026.

Publications (continued)

Collaborative Publications

6. P Khanduri, C Li, **RI Sultan**, A Sarker, Y Qiang, J Kliewer, D Zhu. “On Federated Compositional Optimization: Algorithms, Analysis, and Guarantees.” *TMLR*, 2026.
7. C Li, **RI Sultan**, H Bagher-Ebadian, Y Qiang, K Thind, D Zhu, IJ Chetty. “Enhancing CT Image Segmentation Accuracy Through Ensemble Loss Function Optimization.” *Medical Physics*, 2025.
8. U Mgboh, **RI Sultan**, J Kim, K Thind, D Zhu. “FluenceFormer: Transformer-Driven Multi-Beam Fluence Map Regression for Radiotherapy Planning.” *MIDL*, 2026.
9. C Li, **RI Sultan**, P Khanduri, Y Qiang, C Indrin, D Zhu. “AutoProSAM: Automated Prompting SAM for 3D Multi-Organ Segmentation.” *WACV*, 2025.
10. C Li, H Zhu, **RI Sultan**, HB Ebadian, P Khanduri, C Indrin, K Thind, D Zhu. “MulModSeg: Enhancing Unpaired Multi-Modal Medical Image Segmentation with Modality-Conditioned Text Embedding and Alternating Training.” *WACV*, 2025.
11. C Li, H Bagher-Ebadian, **RI Sultan**, M Elshaikh, B Movsas, D Zhu, IJ Chetty. “A New Architecture Combining Convolutional and Transformer-Based Networks for Automatic 3D Multi-Organ Segmentation on CT Images.” *Medical Physics*, 2023.
12. C Li, Y Qiang, **RI Sultan**, H Bagher-Ebadian, P Khanduri, IJ Chetty, D Zhu. “FocalUNETR: A Focal Transformer for Boundary-Aware Prostate Segmentation Using CT Images.” *MICCAI*, 2023.

Invited Talks & Public Engagement

- “WalkGPT: Pixel-Grounded Navigation Guidance for Pedestrians,” Vox51 Best of CVPR 2026 Series, July 2026.
- “WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation,” Cohere Labs Open Science Community, May 2026.
- “Multimodal AI for Mobility Infrastructure and Pedestrian Guidance,” Wayne State University Departmental Seminar, 2026.
- Featured researcher in “Wayne State University Creates AI for Mobility Project,” *One Detroit*, Detroit PBS, March 2024.

Recognition and Professional Service

- **Outstanding Graduate Research Assistant Award**, Wayne State University Department of Computer Science, 2025.
- **Conference Reviewer/PC**: AAAI 2027; NeurIPS 2026; IJCAI-ECAI 2026; BMVC 2026; IJCNN 2025.
- **Journal Reviewer**: Pattern Recognition; CVIU; IEEE TCSVT; Knowledge-Based Systems; Computer Methods and Programs in Biomedicine; Computers in Biology and Medicine; Biomedical Signal Processing and Control.